

Explicit Process Relations Prevent Semantic Misrouting in Long-Horizon Robotic Wire-Harness Assembly

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DRAFT—NOT FOR SUBMISSION

Abstract—Automotive wire routing requires geometric control and process semantics: cable segments must enter designated fixtures in order. We isolate semantic misrouting through paired interventions that preserve geometry, layout, physics, and seed while changing only the feasible segment–fixture assignment. Our controller represents cable connectivity and process correspondence as typed graph relations and grounds actions with graph pointers. Across 300 paired PBD episodes, the relational policy succeeds in 251 cases (83.7%) versus 10 (3.3%) for an equal-parameter geometry model; paired advantages are 90, 83, and 68 points across three difficulties. All relation-aware models share the same 251/300 outcome, supporting relation availability rather than one graph architecture. In nine-assignment counterfactuals, relation-equipped policies succeed in 2,340/2,700 cells versus 57/2,700 for Geometry. A direct MuJoCo audit gives 180/180 versus 0/180. An offline audit on 1,647 public real-robot cable-routing trajectories confirms a temporal action signal but not semantic success. DLO-Lab distinguishes ordered from wrong-order paths, yet a direct endpoint path scores similarly to the ordered path, exposing limited semantic discrimination. The evidence isolates the need for observable process relations; trained external semantic replication and force-controlled hardware remain open.

Index Terms—Deformable object manipulation, graph representations, industrial robotics, semantic task planning, wire-harness assembly.

I. Introduction

Deformable linear objects (DLOs) combine high-dimensional geometry with contact-rich dynamics. Industrial wire harnesses add a second difficulty: the identities of endpoints, branches, terminals, and fixture-bound segments matter to the manufacturing process. Recent systems have demonstrated contact-aware cable shaping [2], twisting-based harnessing with force control [9], and integrated multi-robot DLO assembly [3]. Concurrent benchmarks now cover industrial clip routing and connector insertion [11], as well as broad DLO manipulation with differentiable physics [1]. Consequently, our contribution is not another claim of being the first industrial DLO simulator.

We instead isolate a semantic misrouting problem. Three cable particles may be physically similar and spatially adjacent, but each is bound to a different clip. Once the wrong segment is latched, low geometric error is not enough to recover the process. This motivates an intervention in which only the segment–clip relation changes. Geometry-only shortcuts should be invariant to

that change, whereas a relational controller should change its decision while preserving performance.

The intended contributions are:

- 1) a paired nine-assignment intervention, frozen by an independent teacher-feasibility screen, that holds the physical RouteBot episode fixed and changes only the segment–clip process relation;
- 2) a typed relational state and graph-pointer policy for long-horizon grasp, transport, latch, and endpoint-placement decisions;
- 3) a controlled comparison with equal-parameter geometry ablation and an independently implemented ACT-style state baseline with direct, typed-pointer, and fixed relational-pooling variants;
- 4) failure-aware statistics, an attributed public real-data audit, and cross-physics replication that distinguish semantic correctness from geometric completion.

II. Related Work

A. Industrial DLO assembly

Contact-aware planning and object-centered skills can maintain DLO shape using fixtures and force feedback [2]. Twisting-induced tension enables a single arm to insert wires into clamps in narrow workspaces [9]. Multi-robot pipelines combine perception, handover, and placement across the assembly workflow [3]. These systems provide stronger real-physics evidence than our present simulation and bound our claims to semantic decision making.

WireCraft spans connector insertion, clip routing, and channel seating with articulated and deformable physics [11]. DLO-Lab emphasizes material diversity, differentiable physics, grasp sensitivity, and long-horizon decomposition [1]. Our experiments target a narrower but orthogonal question: whether a policy respects process relations under exact semantic counterfactuals. External replication on one of these benchmarks is a required final experiment.

Luo et al. collected real Franka Panda demonstrations for single- and multi-clip cable routing and showed that learned recovery and primitive selection matter over long horizons [5]. We use its public LeRobot conversion only as an episode-disjoint offline action-interface audit; the release does not expose our segment–clip relation intervention.

B. Learning DLO dynamics and action sequences

Graph networks have modeled cable deformation for offline-online learning and MPC [8]; graph structure alone is therefore not a new contribution. Our graph edges instead combine physical connectivity with a manufacturing correspondence relation. ACT predicts action chunks with a conditional variational Transformer [10], while Diffusion Policy models multimodal action sequences through conditional denoising [4]. We use a state-based ACT adaptation as the matched temporal baseline and do not claim a comprehensive comparison of every visuomotor sequence model.

III. Problem Formulation

Let $G_t = (V_t, E_t^p, E_t^s)$ denote the observable state at time t . E_t^p contains cable connectivity; E_t^s contains process relations between cable segments, clips, the finish fixture, and the end effector. A route is successful only if every clip c_i is bound to its assigned particle $\pi(i)$ and the endpoint reaches its fixture. Because a crossing in a 2-D projection need not be a topological crossing of the real 3-D harness, we report crossing-free completion as a stricter secondary safety metric rather than mixing it into semantic success. We report geometric coverage separately because $|\{c_i \text{ latched}\}| = 3$ does not imply semantic correctness.

For a physical episode seed s , the counterfactual family selects one cable particle from each of three non-overlapping local arc-length windows. Before the final learned-policy evaluation, we screened nine order-preserving candidates on ten held-out seeds per difficulty. All 270 semantic-completion rollouts succeeded, so we froze all nine assignments. Cable coordinates, clip coordinates, obstacle geometry, material parameters, integration noise, and initial arm state are identical. Only E_t^s and the corresponding success predicate change. This makes each policy comparison paired at the physical-seed level.

IV. Method

A. Typed relational observation

Each node contains normalized position, generic role, endpoint/junction role, process order, activation state, confidence, progress, and difficulty. The policy adds self-loops and performs message passing over $\text{clip}(E^p + E^s, 0, 1)$. Attention pooling is concatenated with global state and a task embedding.

B. Graph-pointer action grounding

Two outputs are learned jointly: continuous action logits and pointer logits over nodes. RouteBot uses one pointer. When no segment is held, the decoder restricts candidates to unbound cable particles and closes the gripper only inside a fixed observable distance. While carrying a segment, candidates are restricted to inactive fixture/finish nodes. The policy never reads hidden success labels.

C. Baselines

The no-topology ablation keeps model parameter count and training schedule fixed while removing adjacency, endpoint/junction identity, process order, and semantic identity. The ACT-style baseline uses a CVAE Transformer, eight-step action chunks, temporal ensembling, and the same typed node observations but no graph edges. We evaluate direct continuous ACT and ACT-Pointer, which shares the candidate filter and learns a node-classification head. The latter controls for the discrete action interface. ACT-RelPool additionally receives the same semantic adjacency but replaces learned graph propagation with fixed one-hop neighbor averaging before its Transformer, controlling for access to the relation itself.

V. Experimental Protocol

The fast backend uses position-based dynamics [6] for paired large-sample experiments. Each main-table cell contains at least 100 unseen seeds at difficulties 0.2, 0.5, and 0.8. All policies receive identical seeds. Binary success uses Wilson intervals; paired differences use McNemar’s exact test and a matched odds ratio. We additionally report clip coverage, topology accuracy, wrong latches, endpoint error, crossings, stretch error, steps, and a mutually exclusive failure taxonomy.

The frozen imitation corpus contains 432 complete teacher trajectories and 19,190 state-action samples (SHA-256 prefix `dac7dd2b8ac`). Thirty training, nine validation, and nine test physical scenes are mutually disjoint. Every physical scene is replayed under all nine assignments, yielding 270/81/81 training/validation/test trajectories; thus assignment identity cannot be confounded with geometry within a scene. The target-particle identity is not a unary feature: changing an assignment changes only semantic adjacency edges and the success predicate.

TopoHarness and its Geometry ablation use identical 64-dimensional, three-layer networks and optimization schedules. Both ACT variants use a 48-dimensional one-layer CVAE Transformer, eight-action chunks, and a two-simulator-step control stride matching demonstration sampling. Model selection uses validation loss; no frozen evaluation seed is used for architecture or assignment selection.

For external real-data grounding, we pin the public Berkeley Cable Routing LeRobot conversion at commit `20a7774a714a5daf56b220e01635425f2cb9745b`. Its 42,328 frames form 1,647 complete real-robot teleoperation episodes with four camera views. We split whole episodes 70/15/15 into 1,153/247/247 train/validation/test episodes, tune ridge regularization on validation episodes only, and report the four non-constant Cartesian action dimensions. A state-only ridge, previous-action control, and state-plus-previous-action ridge are evaluated on the same held-out episodes. The latter is an audit baseline, not our semantic-routing method.

The three-dimensional replication uses MuJoCo [7] cable elasticity. Detailed UR10e and Robotiq meshes are

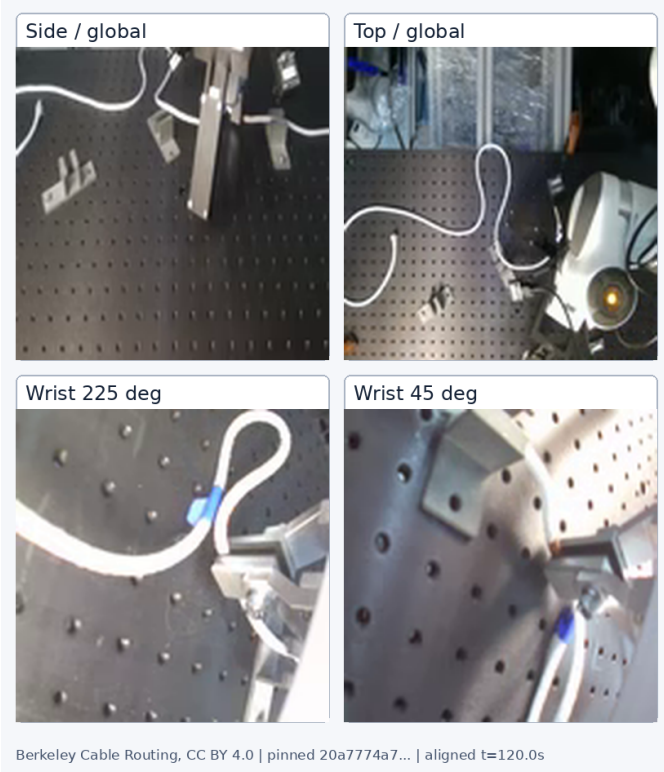


Fig. 1. Aligned views from the pinned public real-robot cable-routing data (Berkeley Cable Routing, CC BY 4.0). Images are evidence of the external data interface, not RouteBot rollouts.

kinematic visual twins with collisions disabled; they must not be interpreted as torque-controlled robot dynamics. For the direct audit, each control step reconstructs the same 29-node typed graph from current MuJoCo cable, fixture, finish, and task-space gripper poses. The frozen network output is mapped back to a MuJoCo mocap target; grasp, latch, and endpoint events are detected solely from native MuJoCo body distances and constraints. No PBD state or event is read. Small paired fixture perturbations define physical seeds, while the nine-assignment intervention changes only semantic graph edges and the success predicate.

For an independently maintained simulator sanity check, we pin DLO-Lab at commit c5026a9 and its prescribed Mushroom-RL fork at ec33647. In the official wiring_post environment, four deterministic open-loop paths run in one matched four-environment batch: no action, a straight endpoint move, a deliberately wrong fixture order, and the target-consistent fixture order. We report DLO-Lab’s native geometric reward without modification, plus ordered curve and fixture-relation diagnostics as an explicitly labeled RouteBot extension. This experiment tests executable cross-simulator plumbing and metric sensitivity, not a trained-policy claim.

VI. Results

a) Frozen paired PBD main table.: TopoHarness completes 251/300 episodes (83.7%, Wilson 95% CI

TABLE I
Paired RouteBot results over all frozen difficulties. Semantic success is primary; crossing-free success is the strict 2-D shape metric.

Policy	Sem. success (%)	Cross.-free (%)
TopoHarness	83.7	73.3
ACT-RelPool	83.7	73.3
Geometry	3.3	2.3
ACT-Pointer	0.0	0.0
ACT direct	0.0	0.0
Semantic-only	83.7	73.3
Physical-only	0.3	0.3
No ID/order feats.	83.7	73.3
TopoHarness (20% data)	83.7	73.3
TopoHarness (50% data)	83.7	73.3
Scripted teacher	100.0	84.7
Random	0.0	0.0

TABLE II
Direct MuJoCo pose-observation/task-space-control audit. The detailed robot is a visual IK twin, not a torque-controlled arm.

Policy	Episodes	Success (%)
Geometry	180	0.0
TopoHarness	180	100.0
Scripted teacher	180	100.0

[79.1, 87.4]), compared with 10/300 for the equal-parameter Geometry model (3.3%, [1.8, 6.0]). The paired success differences at difficulties 0.2, 0.5, and 0.8 are 0.90, 0.83, and 0.68; the corresponding Holm-adjusted exact McNemar values are 5.33×10^{-26} , 5.79×10^{-24} , and 2.29×10^{-18} . Across all difficulties, 242 physical cells succeed only with TopoHarness, one succeeds only with Geometry, nine succeed with both, and 48 fail with both. The scripted teacher succeeds in 300/300 episodes, showing that the residual 49 TopoHarness failures are policy errors rather than an infeasible test set.

The architecture ablation changes the interpretation. ACT-RelPool, Semantic-only, the no-unary-ID/order-feature model, and the 20% and 50% data models all produce exactly the same 251/300 success count and failure partition as TopoHarness. Physical-only reaches 1/300, direct ACT and ACT-Pointer reach 0/300, and random control reaches 0/300. Thus this environment identifies the necessity of an observable process relation and a relation-grounded action interface, but it does not identify learned message passing, unary roles, or the full training-set size as separately necessary. We treat this saturation as a benchmark limitation rather than an architectural win.

b) Frozen direct MuJoCo audit.: Across 20 paired fixture perturbations and all nine segment assignments, TopoHarness and the scripted teacher each complete 180/180 episodes; the equal-parameter geometry policy completes 0/180. The Wilson interval is [0.979, 1.000] for TopoHarness and [0.000, 0.021] for Geometry. All 180 discordant pairs favor TopoHarness (two-sided exact McNemar test with Holm adjustment, $p = 2.61 \times 10^{-54}$).

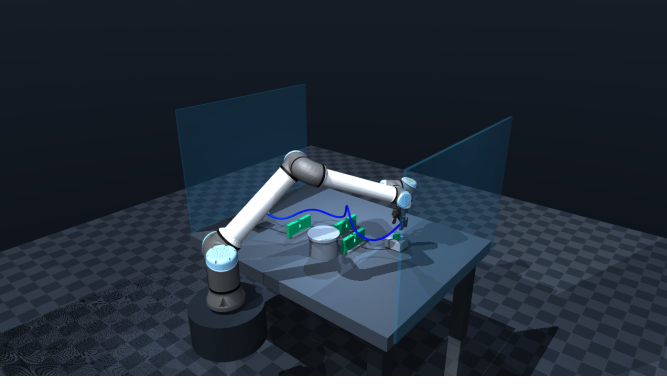


Fig. 2. Frozen TopoHarness final state under direct MuJoCo observation and task-space action control (assignment 7–15–23). The three semantic latches and endpoint succeed; the rendered UR10e is a visual IK twin.

TABLE III

Episode-disjoint offline action prediction on the public real-robot Berkeley Cable Routing data. These are not closed-loop success rates.

Policy	RMSE ↓	NRMSE ↓	Ep. RMSE (95% CI)
Zero	0.2511	1.044	0.2470 (0.2401,0.2538)
Train mean	0.2418	0.997	0.2361 (0.2290,0.2431)
Persistence	0.1719	0.755	0.1711 (0.1654,0.1769)
State ridge	0.1891	0.823	0.1841 (0.1795,0.1893)
State+hist. ridge	0.1536	0.668	0.1507 (0.1459,0.1555)

This isolates relation-sensitive closed-loop decisions under a second cable dynamics backend, but the mocap control abstraction prevents a joint-level or force-control claim.

c) Frozen public real-data audit.: On 247 held-out real-robot episodes, adding the previously executed action to the state ridge reduces mean per-episode active-action RMSE by 11.6% relative to action persistence (paired difference -0.0191 , 95% episode-bootstrap CI $[-0.0206, -0.0176]$). It improves 233 of 247 episodes (two-sided exact sign test $p = 2.33 \times 10^{-52}$). This confirms that the data pipeline detects a real temporal signal; it cannot test semantic binding or closed-loop routing.

TABLE IV

Exact semantic counterfactual over nine assignments. Each policy observes identical physical states within a seed; only relation edges and the corresponding success predicate change.

Policy	Sem. success (%)	Cross.-free (%)
ACT-Pointer	9.9	7.6
ACT-RelPool	86.7	56.3
Geometry	2.1	1.6
TopoHarness	86.7	56.3
Scripted teacher	100.0	65.4

d) Frozen nine-assignment counterfactual.: Across 2,700 physical-seed-assignment cells per policy, TopoHarness succeeds in 2,340 (86.7%, Wilson 95% CI $[85.3, 87.9]$), versus 57 for Geometry (2.1%, $[1.6, 2.7]$), 268 for ACT-Pointer (9.9%, $[8.9, 11.1]$), and 2,700 for the teacher. TopoHarness alone succeeds in 2,284 Geometry-paired cells; Geometry alone succeeds in one, while 56 succeed and 359 fail with both. The aggregate exact McNemar value is 6.41×10^{-685} . More conservatively, each of the 27 assignment-difficulty comparisons has a 60–100 percentage point advantage over Geometry, and the largest Holm-adjusted value is 1.06×10^{-16} . ACT-RelPool again has exactly the same 2,340 successes and failure partition as TopoHarness. Its per-assignment success spans 83–100%, 76–100%, and 60–94% at increasing difficulty; the wider range at high difficulty bounds any claim of perfect assignment invariance.

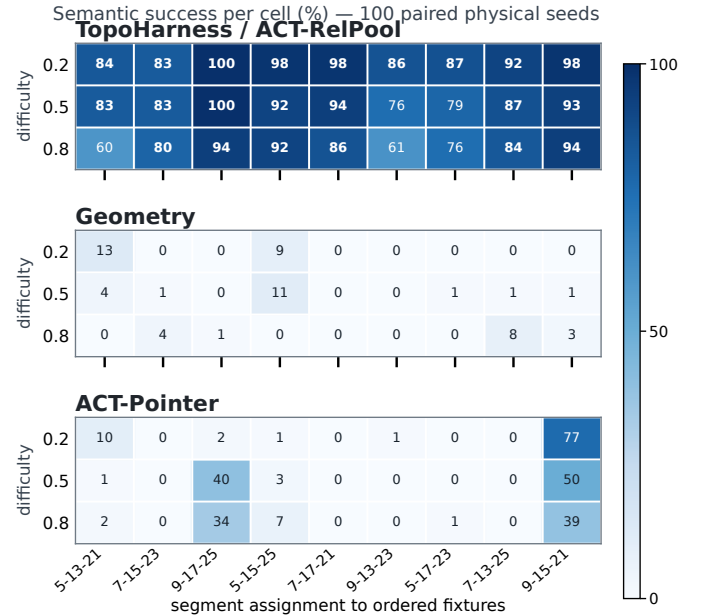


Fig. 3. Cellwise semantic success for the exact relation intervention (100 paired physical seeds per cell; shared 0–100% scale). TopoHarness and ACT-RelPool are cellwise identical. ACT-Pointer succeeds only on isolated assignments, revealing assignment-specific shortcuts.

TABLE V
Matched open-loop sanity check in official DLO-Lab wiring_post.
Native R is unmodified; relation and winding diagnostics are
RouteBot extensions.

Path	Native R $\uparrow$	RMSE (cm) $\downarrow$	Rel. $\uparrow$	Wind. (rad) $\downarrow$
No action	0.089	43.58	0.499	3.05
Endpoint straight	0.970	2.73	0.978	0.10
Wrong fixture order	0.571	12.30	0.750	6.10
Ordered fixture route	0.975	1.46	0.977	0.10

e) Official DLO-Lab execution sanity check.: All four 30-vertex rollouts remain finite. The ordered fixture route reaches native reward 0.975 and ordered-point RMSE 1.46 cm, versus 0.571 and 12.30 cm for the wrong-order path. Its fixture-relation angular error is 0.073 rad, compared with 0.785 rad for wrong order. However, the endpoint-straight path also reaches reward 0.970, 2.73 cm RMSE, and a 0.069-rad relation error. Therefore this result validates the pinned external environment and detects a gross wrong-order path, but the official geometric target does not isolate RouteBot’s semantic intervention. We do not count it as trained external semantic replication.

VII. Limitations and Claim Boundary

The current environments are simulated, the PBD backend is two-dimensional and uncalibrated, and the detailed robot arms do not participate in cable contact. No result in this paper is a production-line success rate. The counterfactual isolates a process relation within a simulator; it does not establish causal effects on real hardware. Force-controlled clip insertion, perception from raw images, cable-parameter identification, and trained external closed-loop semantic replication remain mandatory before a hardware-level claim. The single-seed DLO-Lab matched-path check is open loop and its endpoint-straight control nearly matches the ordered route, so it cannot satisfy that semantic replication requirement. The direct MuJoCo audit uses current three-dimensional object poses but a mocap task-space gripper, so it does not validate contact-force regulation or joint-level control. The public-data experiment is an offline action audit and does not satisfy the external replication gate.

VIII. Ethical and Societal Considerations

The study uses synthetic simulator trajectories and attributed public robot teleoperation traces; we conduct no new human-subject data collection and use no personal data. Wire-harness automation could reduce repetitive ergonomic strain, but it could also displace manual assembly work or intensify monitoring if deployed without worker participation. The present controller is not safety certified: its outputs must not command production robots without guarded workcells, risk assessment, force limits, and qualified human supervision. We therefore report only simulator evidence and publish failure cases as well as successes. The models are small CPU-trainable baselines; training time and hardware are included in the reproducibility record so computational cost is auditable.

IX. Reproducibility

The anonymous package will include episode-disjoint demonstrations, hashes, frozen checkpoints, all seeds and configurations, per-episode CSV files, statistical scripts, MJCF sources, retained third-party licenses, and commands that regenerate every table and figure. The independent ACT adaptation states its differences from the official implementation and does not copy upstream source code. The external DLO-Lab asset archive is not redistributed because it contains no asset license; the package records its official URL, exact hash, required-file hash, and a deterministic local verification command.

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